

Conformal Reliability Routing for Low-Storage Few-Shot Industrial Anomaly Detection

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Abstract—Few-shot industrial anomaly detection increasingly uses frozen vision foundation features, especially DINOv2 patch embeddings with memory-bank or PCA/subspace scoring. These detectors can rank defects well, but their raw distances are not calibrated probabilities and often become unreliable under low-shot, dataset, class, or corruption shift. We present Conformal Reliability Routing (CRR), a decoupled reliability framework for low-storage DINOv2 subspace anomaly detection. CRR keeps PCA/subspace residuals as the raw anomaly ranking score, while a separate reliability layer maps residual evidence to probability-like decision signals through vector calibration and leave-one-image-out (LOIO) conformal p-values computed only from normal support images. This design explicitly separates “which sample is more anomalous” from “how reliable is the alarm probability.” On MVTec AD, the resulting detector is competitive but not state-of-the-art in image AUROC, while requiring substantially less reference storage than memory-bank baselines. On VisA, a PCA128 variant improves AUROC over PCA64 by 0.011–0.017 with storage still below 0.6 MB. On full corruption benchmarks over all 12 VisA and all 15 MVTec classes with $k \in \{4, 8\}$ and four corruptions, LOIO conformal reliability achieves overall ECE of 0.0766 and 0.0684 respectively and improves over Vector Platt and Shift-Aware Platt in every tested corruption cell on both datasets (paired Wilcoxon $p \leq 5 \times 10^{-9}$ except versus Shift-Aware Platt at VisA $k=8$; weighted conformal is a separate ablation, competitive only for selected $k=8$ structured corruptions). Beyond calibration error, we analyze conformal alarms operationally: an attainable-alpha analysis shows that k -shot conformal p-values cannot alarm below $\alpha = 1/(k+1)$, and at the first attainable operating point the target-only rule is conservative on VisA but anti-conservative on corrupted MVTec. To obtain controlled alarms below and at this floor, we introduce SC3R, a source-validated cross-category thresholding scheme that pools normal images from other categories under matched conditions. Against a pre-registered decision gate on all 15 MVTec classes, SC3R tracks nominal false-alarm rates almost exactly (0.050/0.105/0.216 at $\alpha=0.05/0.10/0.20$) with detection power 0.22–0.42 where target-only alarms are structurally silent, and hierarchical class-seed bootstrap intervals for the power gain exclude zero for every corruption condition. The paper positions CRR as a calibrated, low-storage reliability layer for few-shot inspection rather than an MVTec AUROC SOTA or adversarial-robustness claim.

Index Terms—Industrial anomaly detection, few-shot learning, calibration, conformal prediction, DINOv2, reliability, robustness.

I. Introduction

Industrial anomaly detection (AD) is typically deployed under an asymmetric data regime: normal examples are available, while abnormal examples are rare, diverse, and costly to annotate. This makes few-shot AD practically

important. In this setting, an inspection system receives only k normal support images for a target category, often with $k \in \{1, 2, 4, 8\}$, and must detect both image-level and pixel-level defects at test time.

Recent methods have made this regime much more effective by using frozen visual foundation models. DINOv2 [1] patch features can be stored in a normal memory bank and queried by nearest-neighbor distances, as in PatchCore-style methods and AnomalyDINO [2], [3]. Alternatively, frozen patch features can be compressed into a normal PCA/subspace model, as in SubspaceAD [4]. These approaches provide strong anomaly ranking without training a large task-specific backbone.

However, ranking is not the same as reliable decision support. A raw nearest-neighbor distance or PCA residual can produce high AUROC while still being poorly calibrated: a score threshold may work on one class, but the same numerical score may not carry the same false-alarm meaning under another class, dataset, or corruption. This matters in industrial deployment because operators need to know not only which samples are suspicious, but also whether an alarm probability is trustworthy. Prior work has already shown that DINOv2-based few-shot AD can be poorly calibrated and adversarially fragile [5]. Reliability is therefore not a cosmetic post-processing issue; it is part of the deployment problem.

We address this gap by separating anomaly ranking from reliability estimation. The ranking path remains a frozen DINOv2 PCA/subspace detector: patch residuals are aggregated into a raw anomaly score used for AUROC, AP, and localization. A separate reliability path maps this evidence into calibrated or probability-like outputs. We call the resulting framework Conformal Reliability Routing (CRR). The main CRR route uses leave-one-image-out (LOIO) conformal calibration over the few normal support images. Each support image is scored as if held out, producing normal nonconformity values; a test image then receives a conformal p-value and a reliability score $1 - p$. This view is not a supervised anomaly posterior, but it is statistically interpretable and derived without target anomaly labels.

This framing is deliberately conservative. We do not claim that DINOv2 features, PCA residuals, Platt scaling, conformal prediction, or gated experts are individually new. Instead, the contribution is their protocol-locked integration for low-storage few-shot AD: a decoupled

detector whose raw ranking is preserved while reliability is evaluated under clean, transfer, corruption, and representative routing protocols.

The paper makes four contributions:

- We formulate a decoupled few-shot AD pipeline in which frozen DINOv2 PCA/subspace residuals provide anomaly ranking while a separate label-free reliability layer—vector/shift-aware calibration and LOIO conformal p-values [6] computed only from the k normal supports—provides decision signals, with full-benchmark corruption evidence on all 12 VisA and all 15 MVTec classes that the conformal route improves calibration over Platt-family calibrators in every tested cell (paired Wilcoxon tests) while preserving the raw ranking.
- We make the few-shot resolution limit of conformal alarms explicit through an attainable-alpha analysis: with k supports no alarm can fire below $\alpha = 1/(k+1)$, and empirical false-alarm rates must be read against the attainable grid, not the nominal level.
- We introduce SC3R, a source-validated cross-category thresholding scheme that pools normal images from other categories to unlock operating points below the target-only attainable-alpha floor. Under a pre-registered decision gate on all 15 MVTec classes, matched-condition SC3R tracks nominal false-alarm rates almost exactly with substantial power where the target-only anchor is structurally silent, and hierarchical class-seed bootstrap intervals for its power gain exclude zero for every corruption condition.
- We report accuracy-storage, transfer calibration, pixel-level, robustness, prevalence-stress, and routing diagnostics, including negative findings that prevent overclaiming: CRR is not MVTec AUROC SOTA, is not adversarially robust, and ECE on conformal-derived scores is strongly prevalence-sensitive.

II. Related Work

A. Few-Shot Industrial Anomaly Detection

Classical industrial AD often assumes normal-only training data and detects deviations at test time. Patch distribution, self-supervised, and reconstruction-discriminative families such as PaDiM, CutPaste, and DRAEM provide important context for this literature [7]–[9]. PatchCore [2] made memory-bank patch retrieval a strong baseline by storing representative normal features and scoring test patches by nearest-neighbor distance. AnomalyDINO [3] updates this paradigm for few-shot inspection with frozen DINOv2 representations. The few-shot AD literature has also produced registration-based adaptation (RegAD [10]), graph-structured memory (GraphCore [11]), fast feature reconstruction (FastRecon [12]), vision-language zero-/few-shot scoring (WinCLIP [13]), and in-context residual learning (InCTRL [14]). These methods are strong rankers, but the reference state can remain nontrivial at inference time

and the raw score scale is not a calibrated probability; calibration under shift is also known to degrade for deep models generally [15]. Our work is complementary: it does not compete on ranking but asks how any such few-shot detector should expose decision reliability.

B. Frozen-Feature Subspace Detectors

Subspace approaches replace explicit patch retrieval with a compact normal subspace. SubspaceAD [4] is the most relevant recent guardrail: it shows that frozen DINOv2 patch features and PCA/subspace residuals are already strong for few-shot anomaly ranking. Our work builds on this observation rather than claiming it as new. The proposed contribution starts after the residual is computed: how should a low-storage subspace detector expose reliability under low-shot and shift?

C. Calibration and Robustness in Few-Shot AD

Calibration asks whether predicted confidence agrees with empirical correctness [16], [17]. In anomaly detection this is subtle because labels are sparse, anomaly types are open-ended, and the score distribution changes across classes. Reliability analyses of DINOv2-based few-shot AD report poor ECE and strong FGSM fragility [5]. We therefore evaluate ECE, Brier score, and NLL separately from AUROC/AP, and report adversarial results, including FGSM-style perturbations [18], as diagnostics rather than robustness improvements.

D. Conformal Prediction and Reliability Routing

Conformal prediction converts nonconformity scores into p-values under exchangeability assumptions [19]–[21]. This is attractive for normal-only few-shot AD because support normal images can define a reference distribution without requiring real anomalies. Conformal anomaly detection itself is established: leave-one-out, bootstrap, and cross-conformal anomaly detectors have been analyzed directly [6], weighted conformal detection in low-data regimes with resolution and effective-sample-size analyses exists [22], general nonconformity toolkits are available [23], and few-shot conformal prediction with auxiliary tasks addresses the small-calibration-set problem [24]. We therefore do not claim conformal AD, LOIO calibration, or weighted conformal p-values as new mechanisms; our contribution is the operational protocol around them for few-shot inspection, including an explicit attainable-alpha analysis and source-validated cross-category thresholding that operates below the target-only alpha floor without target anomaly labels. Gated-expert models, including SAGE-style shared/specialized routing [25], motivate the broader idea of choosing conservative or shift-specialized reliability paths. CRR uses this routing lens, but its main reported route is intentionally simple: fixed LOIO conformal reliability. Learned or weighted gates are treated as ablations unless they improve reliability without test-label expert selection.

III. Method

A. Problem Setup

For each category c , we observe a support set $\mathcal{S}_c = \{x_i\}_{i=1}^k$ containing only normal images. The test set contains normal and anomalous images, but anomaly labels are used only for evaluation, not for fitting the main detector or calibrator. The goal is to produce: (i) a raw anomaly ranking score $s(x)$, (ii) a probability-like reliability score $\hat{p}(y = 1 | x)$ or conformal anomaly confidence, and (iii) optional pixel-level maps.

B. Frozen Patch Features and Subspace Ranking

We extract patch features with a frozen DINOv2 ViT-S/14 backbone. Let $z_{ij} \in \mathbb{R}^d$ denote patch j from image i . A PCA/subspace model is fit on support patches. With support mean μ and retained principal directions $U \in \mathbb{R}^{d \times m}$, the patch residual is

$$r(z) = \|(z - \mu) - UU^\top(z - \mu)\|_2. \quad (1)$$

Patch residuals form an anomaly map. The image-level ranking score is a high-residual aggregation,

$$s(x) = \text{Agg}_j r(z_j), \quad (2)$$

where Agg is the patch maximum in the clean accuracy-storage benchmark and the mean of the top-1% patch residuals ($\rho=0.01$) in the conformal pipeline; the top-fraction variant is a smoothed maximum that stabilizes the LOIO calibration scores. We state this explicitly because the two protocols use different aggregators. Importantly, this raw score is the quantity used for AUROC and AP. Calibration routes do not change the ranking score.

C. Vector and Shift-Aware Calibration

The first reliability family is post-hoc logistic calibration. A scalar Platt scaler maps $s(x)$ to a probability, while Vector Platt uses a richer descriptor

$$\phi(x) = [s_{\text{pca}}(x), s_{\text{head}}(x), |\tilde{s}_{\text{pca}}(x) - \tilde{s}_{\text{head}}(x)|], \quad (3)$$

where s_{head} is a lightweight synthetic-anomaly head score and tildes denote normalized scores. The head is a small MLP (384→256→1, dropout 0.1) trained with binary cross-entropy to separate support normal patches from synthetically perturbed patches; a patch perturbation resamples a support patch and adds Gaussian noise plus a sign shift in feature space. Its image-level score is the maximum patch probability. The head participates only in the calibration descriptor $\phi(x)$, never in the anomaly ranking. The calibrated probability is

$$\hat{p}_\theta(x) = \sigma(w^\top \phi(x) + b). \quad (4)$$

Shift-Aware Platt augments $\phi(x)$ with five shift descriptors: the maximum and mean distance of test patches to the support feature center (scaled by the support 95th-percentile norm), the mean and standard deviation of patch residual scores, and a residual concentration ratio

(max over mean). Weighted Platt reweights calibration samples by density-ratio estimates from a logistic domain classifier, following weighted conformal prediction under covariate shift [26]. These variants are useful for structured shifts but can over-adapt, motivating a conservative conformal route.

Calibrator training protocol and label source. All calibrators in the main protocol are trained without any real anomaly labels (mode normal_synthetic). The calibration set consists of the k support normal images (label 0) and an equal number of synthetic images (label 1) obtained by replacing a random 25% of a support image’s patch features with perturbed support patches as above. Vector-family calibrators fit a multivariate logistic model on $\phi(x)$ by gradient descent with L_2 regularization; the memory-bank and plain-subspace baselines, which expose no calibration descriptor, use scalar Platt on the raw score under the same synthetic protocol. An anomaly_val mode that calibrates on 10% held-out real anomalies is reported only as an explicitly labeled upper bound. This symmetric, label-free protocol is what makes the LOIO-versus-Platt comparison fair: both families see exactly the same k support normals and no test information.

D. LOIO Conformal Reliability

For $k \geq 2$, CRR constructs normal calibration residuals by leave-one-image-out support splitting. For each support image x_i , a subspace is fit on $\mathcal{S}_c \setminus \{x_i\}$ and the held-out image receives a residual score $s_{-i}(x_i)$. The resulting set

$$\mathcal{R}_{\text{LOIO}} = \{s_{-i}(x_i) : x_i \in \mathcal{S}_c\} \quad (5)$$

serves as normal nonconformity evidence. For a test image x , the conformal p-value is

$$p_{\text{LOIO}}(x) = \frac{1 + \sum_{r \in \mathcal{R}_{\text{LOIO}}} \mathbf{1}\{r \geq s(x)\}}{1 + |\mathcal{R}_{\text{LOIO}}|}. \quad (6)$$

We report $p_{\text{conf}}(x) = 1 - p_{\text{LOIO}}(x)$ as a probability-like anomaly reliability view. It should be interpreted as relative extremeness against held-out support normals, not as a fully supervised posterior probability. Weighted conformal replaces the uniform count by density-ratio weights [26] and reports effective sample size; empirically, it is less stable than LOIO on the main benchmark.

Exchangeability caveat. The construction above is deliberately asymmetric: each calibration residual $s_{-i}(x_i)$ comes from a subspace fit on $k-1$ images, while the test score $s(x)$ uses all k . The two statistics are therefore not exactly exchangeable, and Eq. (6) is a leave-one-out approximation in the spirit of cross-conformal and jackknife+ inference [6], [19], [27] rather than a split-conformal p-value with a formal finite-sample guarantee [28], [29]. We handle this empirically on two fronts. First, a matched-LOIO audit scores held-out support and test images under the same fold-specific subspaces (pairing $s_{-i}(x)$ with r_{-i} , the jackknife+ pairing): it does not improve the operational false-alarm/power trade-off (Section V),

so the simpler construction is retained. Second, rather than assuming validity, we test it directly: Section V reports exact discrete-grid uniformity tests of the normal p-values, which quantify when and in which direction exchangeability fails under corruption shift.

E. Attainable Alpha and the Few-Shot Resolution Floor

Because $|\mathcal{R}_{\text{LOIO}}| = k$, the LOIO p-value is quantized on the grid $\{j/(k+1)\}_{j=1}^{k+1}$ and its smallest attainable value is $1/(k+1)$. Any nominal false-alarm level $\alpha < 1/(k+1)$ therefore cannot raise alarms at all: the detector is not merely conservative but structurally silent. At $k=4$ this floor is 0.2; at $k=8$ it is $1/9 \approx 0.111$. We make this resolution limit explicit with an attainable-alpha analysis that reports, for each nominal α , the nearest attainable grid point and the empirical false-alarm rate achieved there. This turns a commonly observed “few-shot conformal p-values are conservative” effect into an exact, checkable statement, and it defines the regime in which cross-category support is genuinely needed.

F. Source-Conditioned Cross-Category Reliability Routing (SC3R)

The attainable-alpha floor motivates the main routing contribution. In multi-category factories, normal images from other categories (source classes) are plentiful even when the target category has only k normal supports. SC3R uses them in two steps that never touch target anomaly labels. First, target and source residuals are made comparable by support normalization: each raw score is centered and scaled by the robust median/MAD statistics of its own class’s LOIO support residuals. Second, an alarm threshold on conformal p-values is source-validated: for a candidate level α , held-out source classes are conformalized against the remaining source pool, and the largest p-value threshold whose empirical false-alarm rate on held-out source normals stays at or below α is selected. In the matched-condition mode, source images share the target’s corruption condition; in the clean-source mode, only clean source images are used. Pooling hundreds of source normals gives the p-value distribution the resolution that k target supports cannot: operating points below $1/(k+1)$ become attainable, while the source-validation step controls the induced false-alarm rate. A safe-anchor rule keeps target-only LOIO available: at levels at or above the target floor, where the anchor is well defined and strong, SC3R defers to it. The routing component is inspired by the shared-versus-specialized principle in gated-expert systems: a conservative route should remain available, while shift-specialized routes can be used only when they improve reliability without harming the safe baseline. In our implementation, the conservative anchor is LOIO conformal reliability. Candidate experts include Vector Platt, Shift-Aware Platt, weighted Platt, LOIO conformal, and weighted conformal. We evaluate fixed routes, validation-ECE selection, and simple no-label rules

based on shift descriptors and effective sample size. Because learned gates did not yet dominate fixed LOIO across all settings, the main paper reports the fixed LOIO route and treats gated variants as diagnostic ablations.

G. Selective Diagnostics

Beyond producing a calibrated alarm score, CRR records entropy, expert disagreement, conformal confidence, and effective sample size as selective-risk signals. These quantities are not used to change the anomaly ranking. Instead, they support deployment diagnostics such as abstention, risk-coverage analysis, and manual review of low-confidence alarms.

IV. Experiments

A. Datasets and Few-Shot Splits

We evaluate on MVTec AD [30] and VisA [31]. MVTec AD contains industrial object and texture categories with pixel-level defect masks. VisA contains a broader set of industrial categories and is used as the main full-scale corruption and calibration-shift benchmark. For each class, support sets are sampled from normal training images with fixed seeds. Unless otherwise specified, $k \in \{1, 2, 4, 8\}$ is used for clean accuracy-storage studies, and $k \in \{4, 8\}$ is used for the full conformal corruption study because LOIO conformal requires at least two support images and is most relevant in moderate few-shot settings.

B. Evaluation Protocols

We organize experiments into four protocol groups. First, the clean few-shot protocol evaluates image AUROC, AP, calibration, and storage on MVTec and VisA. Second, the accuracy-storage protocol compares PCA64 and PCA128 to test whether a small subspace increase improves accuracy without approaching memory-bank storage. Third, the corruption reliability protocol evaluates Gaussian noise, blur, brightness/contrast shift, and JPEG compression at $k \in \{4, 8\}$ with at most 120 label-stratified images per case: on VisA across all 12 classes and five seeds (480 cases, 56,000 image rows), and on MVTec across all 15 classes and three seeds (360 cases, 37,464 image rows). Fourth, representative routing protocols evaluate MVTec→VisA, VisA→MVTec, leave-one-class-out (LOCO), and within-dataset class splits. Fifth, the operational false-alarm protocol thresholds conformal p-values at nominal levels $\alpha \in \{0.01, 0.05, 0.10, 0.20\}$ and reports false-alarm rate on normal images, detection power, alarm precision, and the attainable-alpha structure; the SC3R protocol additionally selects thresholds on held-out source-class normals with label-stratified random image sampling and a fixed sampling manifest, and is judged against a decision gate registered before the stratified run (nonzero power at $\alpha \leq 0.10$, mean false-alarm rate at most $\alpha+0.02$, no-harm at least 80%, no target anomaly labels, hierarchical class-seed bootstrap support).

C. Naming Convention

CRR denotes the overall framework (decoupled ranking plus reliability layer). The detector instantiations are named by their subspace configuration: CRR/PCA64 (default, also called CalibSubspaceHead in configuration files) and CRR/PCA128. Reliability views are named by their route: scalar/Vector/Shift-Aware/weighted Platt, LOIO conformal, and weighted conformal. Tables use these names consistently.

D. Baselines and Ablations

We compare against a controlled DINOv2 nearest-neighbor memory-bank baseline, a SubspaceAD-style PCA residual baseline, scalar Platt, Vector Platt, Shift-Aware Platt, weighted Platt, LOIO conformal, and weighted conformal. The controlled memory-bank implementation is not an official reproduction of PatchCore or AnomalyDINO; it holds the frozen DINOv2 feature pipeline and data split fixed to isolate the scoring and storage trade-off. In addition, we run the official AnomalyDINO code (ViT-S/14 at 448px, agnostic preprocessing) on MVTec for $k \in \{1, 4, 8\}$ with three seeds and report it as a separate accuracy reference. Official SubspaceAD representative runs are included as a novelty guardrail. Their strong performance confirms that frozen DINOv2 subspace residuals are an existing competitive ranking mechanism; CRR should therefore be judged by reliability, storage, and diagnostic value rather than by claiming the residual itself as novel.

E. Metrics

We separate ranking metrics from reliability metrics. AUROC and AP are computed from the raw PCA/subspace residual score $s(x)$. Pixel-level localization is measured by pixel AUROC and AU-PRO when masks are available. Reliability is measured by expected calibration error (ECE), Brier score, and negative log-likelihood (NLL) using the selected reliability probability or probability-like conformal view. Storage is measured as retained reference/model state per class. Robustness experiments include corruption shift and a label-aware PCA-residual FGSM surrogate; the latter is used only as a fragility diagnostic.

F. Reproducibility and Claim Rules

All runs are indexed by dataset, class, k , seed, model variant, calibration mode, and corruption/attack setting. The core rule is that reliability routes cannot claim ranking improvement unless the raw score changes. Conversely, ranking methods cannot claim calibrated probabilities unless ECE/Brier/NLL improve. This separation prevents a common ambiguity in AD papers: conflating a better sorted list with a more trustworthy alarm probability.

TABLE I

Clean image-level performance and storage. Best accuracy within each dataset/ k block is bold; CRR-style subspace models emphasize low storage rather than universal AUROC dominance. Official AnomalyDINO rows are reproductions with the released code (ViT-S/14, 448px, agnostic preprocessing, 3 seeds) and use its own evaluation protocol; ECE and per-class storage are not measured under our audit and are left blank. Controlled-NN ECE values are diagnostic, not competitive: under the label-free scalar-Platt protocol this baseline’s probability collapses to a constant (≈ 1 while all support patches self-match at $k \leq 2$; ≈ 0 once the 4096-patch memory cap forces subsampling at $k \geq 4$), so ECE equals a prevalence complement (0.277/0.693) and storage saturates at exactly 6.000 MB. CRR ECE at $k=1$ comes from the vector-Platt view, since LOIO conformal requires $k \geq 2$. All rows are means over classes and 5 seeds.

Dataset	k	Method	AUROC $\uparrow$	AP $\uparrow$	ECE $\downarrow$	MB $\downarrow$
MVTec	1	Official AnomalyDINO	0.965	0.982	–	–
		Controlled DINOv2 NN	0.914	0.952	0.277	2.005
		CRR / CalibSubspace	0.904	0.951	0.274	0.472
	4	Official AnomalyDINO	0.976	0.984	–	–
		Controlled DINOv2 NN	0.942	0.967	0.693	6.000
		CRR / CalibSubspace	0.937	0.967	0.214	0.472
	8	Official AnomalyDINO	0.980	0.990	–	–
		Controlled DINOv2 NN	0.948	0.970	0.693	6.000
		CRR / CalibSubspace	0.945	0.972	0.154	0.472
VisA	1	Controlled DINOv2 NN	0.804	0.809	0.434	2.005
		CRR / PCA64	0.823	0.834	0.429	0.472
		CRR / PCA128	0.834	0.842	0.430	0.566
	4	Controlled DINOv2 NN	0.862	0.861	0.543	6.000
		CRR / PCA64	0.870	0.883	0.284	0.472
		CRR / PCA128	0.885	0.895	0.278	0.566
	8	Controlled DINOv2 NN	0.873	0.870	0.533	6.000
		CRR / PCA64	0.880	0.891	0.205	0.472
		CRR / PCA128	0.897	0.905	0.170	0.566

V. Results

A. Clean Accuracy and Storage

Table I summarizes clean image-level performance and storage. The official AnomalyDINO reproduction (released code, 3 seeds) leads MVTec image AUROC at every k (0.965/0.976/0.980), which we report as an explicit accuracy reference: neither the controlled nearest-neighbor baseline nor CRR claims ranking superiority on MVTec. CalibSubspaceHead is competitive with the controlled DINOv2 nearest-neighbor memory-bank baseline while being far more compact: PCA64 stores about 0.472 MB, compared with roughly 2–6 MB for the controlled memory bank. On VisA, the detector is stronger relative to this controlled baseline, and the PCA128 ablation improves VisA AUROC by 0.011–0.017 over PCA64 while keeping storage below 0.6 MB. The controlled rows must not be read as official PatchCore or AnomalyDINO reproductions; only the official AnomalyDINO rows are.

Storage accounting and compute cost. The per-class figures above cover only the class-specific reference state and deliberately exclude the shared frozen DINOv2 ViT-S/14 backbone (≈ 22 M parameters, ≈ 84 MB in fp32), which every compared method requires once per deployment. The 0.472 MB CRR state decomposes into the PCA subspace

TABLE II

Full VisA conformal reliability under corruptions. The PCA/subspace ranking score is fixed; only the reliability view changes. Best reliability values within each group are bold.

Group	Reliability view	N	AUROC	AP	ECE ↓	Brier ↓	NLL ↓
All	LOIO conformal	56000	0.823	0.846	0.077	0.187	0.686
All	Weighted conformal	56000	0.823	0.846	0.166	0.244	0.813
$k=4$	LOIO conformal	28000	0.820	0.842	0.039	0.187	0.760
$k=4$	Weighted conformal	28000	0.820	0.842	0.216	0.259	0.925
$k=8$	LOIO conformal	28000	0.828	0.851	0.114	0.188	0.612
$k=8$	Weighted conformal	28000	0.828	0.851	0.116	0.229	0.701

TABLE III

ECE comparison on the full VisA corruption benchmark. Lower is better. Bold indicates the best reliability view in each corruption cell.

k	Corruption	Vector	Shift-aware	LOIO	Weighted
4	blur	0.284	0.211	0.037	0.197
4	brightness/contrast	0.285	0.212	0.036	0.199
4	gaussian noise	0.270	0.276	0.040	0.256
4	jpeg	0.288	0.230	0.043	0.210
8	blur	0.208	0.144	0.108	0.089
8	brightness/contrast	0.209	0.153	0.109	0.093
8	gaussian noise	0.190	0.191	0.117	0.189
8	jpeg	0.212	0.156	0.123	0.099

(64 components $\times$ 384 dims in fp32 $\approx$ 0.098 MB, matching the observed 0.094 MB PCA128–PCA64 increment) and the synthetic-anomaly head and calibrator ($\approx$ 0.37 MB); the controlled memory bank stores up to 4096 patch vectors ($\leq$ 6.0 MB). At inference, cached-feature scoring costs about 1.3 ms per image for the subspace detector versus 5–13 ms for the nearest-neighbor bank, while the shared backbone forward dominates end-to-end latency (order of seconds per batch on first pass, identical across methods). Fitting the k leave-one-out subspaces for LOIO adds a one-time per-class setup cost of k PCA fits on at most $k \times 1369$ patch vectors, i.e., well under a second at $k \leq 8$.

B. Full VisA Conformal Reliability

Table II gives the main calibration-error result. Throughout the paper, ECE uses 15 equal-width bins; because few-shot conformal confidences occupy at most $k+1$ distinct values, we also report an equal-mass adaptive ECE below, and Section V-D replaces ECE with conformal-native diagnostics for the central claim. Across 56,000 full VisA corruption rows, LOIO conformal obtains pooled ECE 0.0766 while preserving the same PCA/subspace ranking. The strongest regime is $k = 4$, where pooled ECE drops to 0.0391. At $k = 8$, ECE increases to 0.1140 but remains better than Platt-style baselines in every corruption cell. The uniformity analysis in Section V-D explains this k trend mechanistically: with a clean support and corrupted test images, normal test residuals are inflated relative to the clean LOIO calibration set; the tighter $k=8$ subspace amplifies this mismatch, pushing normal p-values anti-conservative.

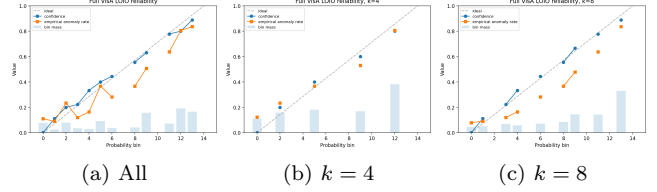


Fig. 1. Reliability diagrams for LOIO conformal reliability on full VisA corruptions. The $k = 4$ setting is especially well calibrated; $k = 8$ remains improved over Platt baselines but shows higher normal-sample confidence.

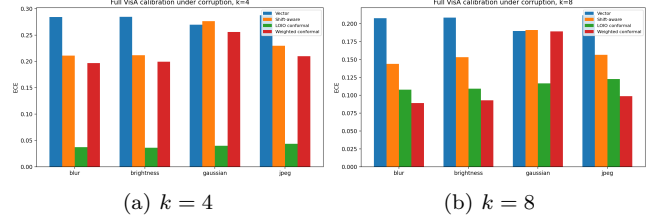


Fig. 2. ECE under full VisA corruptions. LOIO conformal is the most stable main reliability view; weighted conformal is beneficial only in selected $k = 8$ structured corruptions.

Table III isolates corruption effects. LOIO conformal improves over Vector Platt and Shift-Aware Platt for blur, brightness/contrast shift, Gaussian noise, and JPEG at both $k = 4$ and $k = 8$. Weighted conformal is a useful ablation but not a safe default: it is competitive for selected $k = 8$ structured corruptions, yet it is much worse at $k = 4$ and under Gaussian noise.

C. Full MVTEC Conformal Reliability

The same conformal protocol was run on all 15 MVTEC classes ($k \in \{4, 8\}$, seeds 0–2, four corruptions, 37,464 image rows). The VisA conclusion replicates: LOIO conformal reliability attains an overall ECE of 0.0684 (0.0600 at $k=4$, 0.0769 at $k=8$) while preserving the raw ranking (image AUROC 0.912), and it improves ECE over both Vector Platt and Shift-Aware Platt in every one of the eight k -corruption cells, often by a factor of two to four (e.g., $k=4$ Gaussian noise: 0.033 versus 0.233 and 0.270). Weighted conformal is again markedly worse (ECE 0.31–0.45). Conformal reliability is therefore not a VisA-specific effect; it holds across both benchmarks under identical protocols.

D. Statistical Validity: Uniformity Tests, Paired Significance, and Adaptive ECE

Conformal p-values promise $P(p \leq \alpha) \leq \alpha$ on normal data under exchangeability—a false-alarm guarantee, not a posterior probability. We therefore test this property directly instead of relying on ECE. Because a k -shot LOIO p-value is discrete on the grid $\{j/(k+1)\}$, a continuous Kolmogorov–Smirnov test against $U(0, 1)$ is invalid by construction; we instead compare the empirical CDF of

normal p-values to its exact expectation at every grid point and calibrate the resulting discrete KS statistic against the exact null by Monte Carlo. Uniformity is rejected in all 16 dataset- k -corruption cells (Monte Carlo $p \leq 0.005$), but the direction of the violation is the informative result. On VisA at $k=4$ the deviation is uniformly conservative (empirical CDF below nominal; worst gap -0.10 at the 0.4 grid point), consistent with the below-nominal false-alarm rates above. On VisA at $k=8$ and on MVTec at both k , the deviation flips anti-conservative under corruption—most severely for Gaussian noise on MVTec $k=4$, where $\hat{F}(0.2) = 0.46$ against a nominal 0.20. This is exactly the clean-support/corrupted-test mechanism anticipated by the exchangeability caveat in the method section: corruption inflates normal test residuals relative to the clean LOIO calibration set, and a tighter $k=8$ subspace amplifies the mismatch. A matched-LOIO audit that scores support and test under the same fold-specific subspaces (the jackknife+ pairing [27]) does not repair this trade-off: on representative MVTec $k=4$ it raises clean false alarms from 0.17 to 0.25 with essentially unchanged power. The violation is driven by the support-test condition mismatch, not by the leave-one-out asymmetry alone—which is precisely the failure mode SC3R’s matched-condition source validation addresses.

Paired significance replaces point estimates for the head-line calibration comparison. Over class-seed-corruption cells, LOIO conformal beats Vector Platt with Wilcoxon signed-rank $p \leq 5 \times 10^{-9}$ on both benchmarks and both k (MVTec $k=4$: 0.120 ± 0.054 versus 0.223 ± 0.095 , LOIO better in 84% of 360 cells; VisA $k=4$: 0.131 ± 0.064 versus 0.286 ± 0.060 , better in 97%). Against Shift-Aware Platt the result holds everywhere except VisA $k=8$, where the difference is not significant ($p=0.20$, LOIO better in 51% of cells)—an honest boundary we state rather than average away. Replacing equal-width with equal-mass (adaptive) ECE, which cannot be flattered by at most $k+1$ occupied confidence levels, increases LOIO’s per-cell ECE (MVTec $k=4$: $0.120 \rightarrow 0.160$; VisA $k=4$: $0.132 \rightarrow 0.227$) while leaving the ordering against Platt baselines intact. Binning does inflate the apparent margin, which is one more reason the operational analyses below, not ECE, carry the central claim.

E. Attainable Alpha and Operational False-Alarm Control

Calibration error alone does not answer the operator’s question: “if I alarm at $p \leq \alpha$, how often is a good part flagged?” Table IV reports the attainable-alpha analysis. The pattern is structural, not empirical: at $k=4$ every nominal level below 0.2 raises zero alarms on both datasets because no attainable p-value exists below the floor $1/(k+1)$, and at $k=8$ the same holds below $1/9$. Reporting these zeros as “conservative coverage” would be misleading; they are resolution artifacts of the few-shot calibration set.

TABLE IV
Attainable-alpha structure of few-shot LOIO conformal p-values. With k support scores the smallest attainable p-value is $1/(k+1)$, so nominal levels below this floor structurally raise no alarms; empirical false-alarm rates (FAR, mean over classes/seeds/corruptions) then sit at the nearest attainable grid point rather than at the nominal level.

dataset	k	nominal α	floor	attainable	FAR	detection
VisA (full)	4	0.01	0.200	0.000	0.000	0.000
	4	0.05	0.200	0.000	0.000	0.000
	4	0.10	0.200	0.000	0.000	0.000
	4	0.20	0.200	0.200	0.152	0.599
	8	0.01	0.111	0.000	0.000	0.000
	8	0.05	0.111	0.000	0.000	0.000
	8	0.10	0.111	0.000	0.000	0.000
	8	0.20	0.111	0.111	0.111	0.537
MVTec (full)	4	0.01	0.200	0.000	0.000	0.000
	4	0.05	0.200	0.000	0.000	0.000
	4	0.10	0.200	0.000	0.000	0.000
	4	0.20	0.200	0.200	0.411	0.887
	8	0.01	0.111	0.000	0.000	0.000
	8	0.05	0.111	0.000	0.000	0.000
	8	0.10	0.111	0.000	0.000	0.000
	8	0.20	0.111	0.111	0.304	0.907

TABLE V
Operational false-alarm behavior of target-only LOIO conformal p-values at the first attainable operating point ($\alpha=0.20$; alarms fire at $p \leq \alpha$ with a float tolerance). Rates are pooled over classes and seeds. On VisA the rule is conservative (FAR below nominal); on full MVTec it is anti-conservative under corruption, especially at $k=4$, motivating source-validated thresholding (Table VI).

dataset	corruption	k	FAR	detection	precision
VisA (full)	blur	4	0.155	0.610	0.806
	blur	8	0.109	0.546	0.841
	bright/contr.	4	0.155	0.595	0.803
	bright/contr.	8	0.110	0.533	0.836
	Gauss. noise	4	0.142	0.585	0.813
	Gauss. noise	8	0.103	0.522	0.843
	JPEG	4	0.157	0.604	0.803
	JPEG	8	0.123	0.547	0.825
MVTec (full)	blur	4	0.306	0.887	0.872
	blur	8	0.194	0.908	0.916
	bright/contr.	4	0.313	0.887	0.869
	bright/contr.	8	0.188	0.908	0.919
	Gauss. noise	4	0.463	0.879	0.816
	Gauss. noise	8	0.355	0.898	0.855
	JPEG	4	0.410	0.890	0.836
	JPEG	8	0.307	0.914	0.875

Table V evaluates the first attainable operating point, $\alpha=0.20$, on both full benchmarks. On VisA the LOIO alarm rule is conservative but functional (VisA $k=4$: false-alarm 0.14–0.16 with detection 0.58–0.61 and precision around 0.81); on full MVTec the same rule is anti-conservative under corruption at $k=4$, which the SC3R analysis below addresses directly. We note a practical pitfall uncovered during this analysis: p-values stored in single precision can exceed their exact rational value (e.g., $1/5 \rightarrow 0.20000000298$), so an exact comparison $p \leq \alpha$ silently discards every alarm at the floor. All thresholding in this paper uses a small tolerance, and we recommend

TABLE VI

Source-validated thresholding (SC3R, matched-condition source mode) versus the target-only LOIO anchor on all 15 label-stratified MVTec classes ($k=4$, seeds 0–2). Below the target-only attainable-alpha floor $1/(k+1)=0.2$, the anchor cannot raise alarms; source pooling unlocks these operating points with controlled false alarms. Bold marks cells satisfying $\text{FAR} \leq \alpha+0.02$ with nonzero power.

corruption	α	SC3R (source-validated)			Target-only anchor		
		FAR	power	prec.	FAR	power	prec.
clean	0.05	0.050	0.267	0.928	0.000	0.000	–
	0.10	0.100	0.470	0.957	0.000	0.000	–
	0.20	0.209	0.768	0.914	0.267	0.863	0.898
blur	0.05	0.048	0.242	0.915	0.000	0.000	–
	0.10	0.099	0.468	0.950	0.000	0.000	–
	0.20	0.207	0.760	0.904	0.269	0.860	0.895
bright/contr.	0.05	0.050	0.254	0.914	0.000	0.000	–
	0.10	0.101	0.467	0.954	0.000	0.000	–
	0.20	0.205	0.769	0.905	0.268	0.863	0.898
Gauss. noise	0.05	0.048	0.184	0.934	0.000	0.000	–
	0.10	0.105	0.295	0.929	0.000	0.000	–
	0.20	0.226	0.495	0.914	0.460	0.860	0.844
JPEG	0.05	0.051	0.134	0.911	0.000	0.000	–
	0.10	0.121	0.381	0.936	0.000	0.000	–
	0.20	0.235	0.633	0.917	0.434	0.867	0.854

the same in deployments.

F. SC3R: Operating Below the Target-Only Alpha Floor

Table VI evaluates SC3R against the target-only LOIO anchor on all 15 label-stratified MVTec classes ($k=4$, seeds 0–2, five conditions, 675 evaluation cells), under a pre-registered decision gate fixed before the run: nonzero power at $\alpha \in \{0.05, 0.10\}$, mean false-alarm rate at most $\alpha+0.02$, no-harm versus the anchor in at least 80% of class–seed–corruption cells, no target anomaly labels, and hierarchical class–seed bootstrap support. Matched-condition SC3R passes every criterion. It achieves mean power 0.216 at $\alpha=0.05$ and 0.416 at $\alpha=0.10$ in a regime where the target-only anchor is structurally silent, while mean false-alarm rates track nominal almost exactly: 0.050, 0.105, and 0.216 at $\alpha \in \{0.05, 0.10, 0.20\}$, with per-condition false-alarm rates between 0.048 and 0.051 at $\alpha=0.05$ and alarm precision above 0.90 throughout. No-harm rates are 89%, 82%, and 89%. The hierarchical class–seed bootstrap is decisive at this scale: the 95% confidence interval for the power gain over the anchor excludes zero for every condition at both sub-floor levels (e.g., clean $\alpha=0.10$: [0.30, 0.64]; JPEG $\alpha=0.05$: [0.02, 0.28]). At $\alpha=0.20$, at the anchor’s floor, the roles reverse in an instructive way: the anchor attains higher power but its false-alarm rate inflates to 0.27–0.46 under corruption, well above nominal, whereas SC3R remains within 0.005–0.035 of nominal. Source validation thus buys consistent false-alarm tracking at every level, at the cost of some power at levels the anchor can reach. Residual weaknesses are small and localized: JPEG at $\alpha=0.10$ (0.121) and Gaussian noise/JPEG at $\alpha=0.20$ (0.226/0.235) exceed the $\alpha+0.02$

TABLE VII

Protocol-locked representative routing. Fixed LOIO conformal does not require validation-label expert selection and reduces ECE relative to Vector Platt.

Protocol	Vector ECE	LOIO ECE	Δ ECE	No-harm
MVTec→VisA	0.301	0.099	-0.203	1.000
VisA→MVTec	0.304	0.102	-0.202	1.000
LOCO	0.304	0.116	-0.188	1.000
Within split	0.361	0.154	-0.207	1.000

TABLE VIII

Additional diagnostic evidence. Pixel ranking is competitive, while FGSM remains a fragility diagnostic rather than a robustness claim.

Setting	Method/View	k	Value
MVTec pixel AUROC	CRR / CalibSubspace	1	0.944
MVTec pixel AUROC	Controlled DINOv2 NN	1	0.940
MVTec AU-PRO	CRR / CalibSubspace	1	0.820
MVTec pixel AUROC	CRR / CalibSubspace	8	0.960
MVTec pixel AUROC	Controlled DINOv2 NN	8	0.952
MVTec AU-PRO	CRR / CalibSubspace	8	0.876
FGSM diagnostic	PCA-residual surrogate	all	severe collapse
Selective risk	entropy/conformal signals	all	diagnostic only

budget per-condition while the pooled means stay within it.

G. ECE Is Prevalence-Sensitive for Conformal Scores

A further protocol warning emerged from prevalence stress tests: ECE computed on $1 - p$ conformal scores changes drastically with the anomaly prevalence of the evaluation set even when the ranking is fixed. On full VisA, LOIO ECE moves from 0.404 at 1% prevalence to 0.149 at 50%, and the LOIO-versus-weighted ranking reverses across this range. ECE on anomaly-balanced test sets therefore cannot certify deployment reliability of conformal-derived scores. We report ECE as a secondary metric only and rely on false-alarm rate, power, precision, and risk-coverage for operational claims. The risk-coverage view also quantifies the value of abstention: ranking images by predictive entropy and abstaining on the most uncertain 20% halves the remaining LOIO calibration error on full MVTec (0.068 to 0.030) and improves it moderately on VisA (0.077 to 0.075 at 80% coverage, 0.066 at 70%), so selective operation is a practical complement to fixed- α alarms.

H. Routing, Pixel Localization, and Fragility Diagnostics

Table VII evaluates protocol-locked representative routing. Fixed LOIO conformal reduces ECE relative to Vector Platt without using validation labels to select an expert. This supports the central CRR claim: reliability routing can be useful even when the route is fixed and conservative. SAGE-style gating remains a promising design inspiration, but the current main evidence favors simple LOIO routing over learned or weighted expert selection.

Pixel-level results show that the low-storage subspace detector is also competitive for localization. Robustness

TABLE IX

Novelty positioning. The paper claims reliability routing around a low-storage subspace detector, not novelty of each ingredient in isolation.

Work family	Already covers	Our safe claim boundary
AnomalyDINO	DINOv2 few-shot patch/memory-bank AD	Not DINOv2 few-shot novelty
SubspaceAD	DINOv2 PCA/subspace residual ranking	Not PCA residual novelty
Calibration/robustness studies	ECE, Platt scaling, FGSM fragility	Not first calibration/FGSM benchmark
Conformal AD	P-values from nonconformity scores	Not first conformal AD
SAGE-style routing	Gated shared/specialized experts	Routing inspiration only

diagnostics are less favorable: FGSM experiments show severe fragility and objective sensitivity. These results are important negative evidence and motivate reliability diagnostics, but they do not support an adversarial robustness claim.

I. Novelty Guardrails

Table IX summarizes the claim boundary. The official SubspaceAD representative run is especially important because it reaches high image and pixel AUROC in our protocol. The paper should therefore be positioned around calibrated low-storage reliability and conformal diagnostics, not around inventing DINOv2 PCA residual ranking.

VI. Limitations and Next Experiments

CRR is intended as a reliability framework, not as a universal accuracy replacement. First, the method is not an MVTEC AUROC SOTA claim. Official SubspaceAD remains a very strong ranking baseline, and our MVTEC contribution is mainly storage, calibration, and diagnostics. Second, CRR is not adversarially robust: label-aware PCA-residual FGSM tests show severe collapse and objective sensitivity. Third, conformal reliability depends on whether the few normal support images are representative of future normal data. If support normals are not exchangeable with test normals, p-values can become overconfident. Fourth, SC3R’s evidence has boundaries: the gate-passing result covers all 15 MVTEC classes but only $k=4$ and seeds 0–2, per-condition false-alarm rates slightly exceed the $\alpha+0.02$ budget for JPEG at $\alpha=0.10$ (0.121) and for Gaussian noise/JPEG at $\alpha=0.20$ (0.226/0.235), it requires a multi-category deployment with matched acquisition conditions, and cross-dataset transfer of source archives is untested. Fifth, ECE on conformal-derived scores is strongly prevalence-sensitive, so calibration numbers on anomaly-balanced test sets must not be read as deployment reliability; we treat operational false-alarm, power, and precision metrics as primary.

The remaining extensions are scoped: SC3R at $k=8$ and with more seeds, cross-dataset source archives (e.g., MVTEC sources for VisA targets), randomized/smoothed few-shot p-values as an alternative route below the attainable-alpha floor, temperature scaling, isotonic regression, and histogram binning as additional label-free calibration baselines under the same synthetic protocol,

and a vision–language few-shot baseline such as WinCLIP [13] run under our audit rather than cited from reported numbers.

VII. Conclusion

We introduced Conformal Reliability Routing, a decoupled framework for low-storage few-shot industrial anomaly detection. CRR preserves frozen DINOv2 PCA/subspace residuals for anomaly ranking and adds separate calibration and conformal routes for probability-style reliability. The main empirical results are threefold: LOIO conformal reliability substantially reduces ECE on full VisA and full MVTEC corruption benchmarks (overall ECE 0.0766 and 0.0684) while leaving the raw ranking unchanged; an attainable-alpha analysis makes the structural silence of k -shot conformal alarms below $1/(k+1)$ explicit and shows that the first attainable target-only operating point is conservative on VisA but anti-conservative on corrupted MVTEC; and SC3R, a source-validated cross-category thresholding scheme, passes a pre-registered gate on all 15 MVTEC classes, tracking nominal false-alarm rates almost exactly below and at the floor without target anomaly labels, with CI-supported power gains under every corruption condition. The resulting story is deliberately focused: CRR is not an MVTEC AUROC SOTA or adversarial-robustness method; it is a calibrated, storage-light reliability layer for few-shot inspection systems where raw anomaly scores alone are not sufficient for deployment-quality decisions.

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